

A Conditional Composite Exterior Route

Liminf Closure and the Minimal Remaining Physical Hypotheses

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Abstract

We assemble the finite-memory exterior route into a single conditional all-level theorem. For each level, four independently valid lower candidates for the relative fourth singular mode are admitted: direct Weyl, spectral four-volume divided by three-mode capacity, normalized exterior trace divided by concentration and capacity, and a directional Rayleigh volume corrected by a relative tail factor. Their maximum is a lower bound, and a strict liminf crossing of the target threshold implies eventual support even when the winning route alternates with level. Route-specific outward transport losses enter only by downward degradation. The aligned five-scale audit has 360 threshold-labelled records. The RH-115 composite closes 321 of 322 genuinely supported records, while the internally consistent RH-116 adaptive route closes all 322; both have zero false positives and their support labels agree. The theorem isolates three minimal physical condition packets—direct margin, trace–concentration, and directional Rayleigh. Their finite algebra and audits are closed, but none of their all-level liminf conditions is proved. The result is a route map, not an unconditional Stage A or Riemann Hypothesis theorem.

1 Factor ledger at one level

Let K be a nonzero projected cross with singular ratios

$$q_j = \frac{s_j(K)}{s_1(K)}, \quad \nu_4 = q_2 q_3 q_4, \quad \Lambda_{23} = q_2 q_3.$$

The exact identity is

$$q_4 = \frac{\nu_4}{\Lambda_{23}}. \tag{1}$$

Assume all numerical quantities below enclose this same operator K , or have first been transported to it by an outward perturbation radius in the sense of interval analysis [Moore, 1966].

Let D^- be any direct lower for q_4 , let V^- be any lower for ν_4 , and let $L^+ > 0$ satisfy $\Lambda_{23} \leq L^+$. Then $V^-/L^+ \leq q_4$ by (1).

The trace route can be written intrinsically. Define

$$\Theta_4(K) = \frac{e_4(s(K)^2)}{s_1(K)^8}, \quad \kappa_4(K) = \frac{e_4(s(K)^2)}{s_1(K)^2 s_2(K)^2 s_3(K)^2 s_4(K)^2}.$$

Whenever the denominator is nonzero,

$$\nu_4(K)^2 = \frac{\Theta_4(K)}{\kappa_4(K)}. \quad (2)$$

Thus $\Theta^- \leq \Theta_4$ and $\kappa_4 \leq \kappa^+$ give

$$q_4 \geq \frac{\sqrt{\Theta^-/\kappa^+}}{L^+}.$$

RH-111 provides the finite-memory mechanism for constructing these one-sided inputs.

For the directional route, let A be a four-column recent-frame action and let the residual action satisfy the relative Gram inequality

$$R^*R \preceq (\gamma^+)^2 A^*A.$$

The RH-114 determinant comparison gives a multiplicative factor $(1 - \gamma^+)_+^4$. If a^- is a normalized lower for the recent-frame four-volume, then

$$q_4 \geq \frac{(1 - \gamma^+)_+^4 a^-}{L^+}.$$

The matrix inequalities used in these conversions are standard singular and Rayleigh comparison principles [Horn and Johnson, 1991].

2 The conditional all-level gate

Index levels by n . At level n , assume admitted one-sided data

$$D_n^-, \quad V_n^-, \quad \Theta_n^-, \quad \kappa_n^+, \quad a_n^-, \quad \gamma_n^+, \text{quad} L_n^+.$$

Zero is used whenever a route lacks a valid input. Define

$$B_n = \max \left\{ D_n^-, \frac{V_n^-}{L_n^+}, \frac{\sqrt{\Theta_n^-/\kappa_n^+}}{L_n^+}, \frac{(1 - \gamma_n^+)_+^4 a_n^-}{L_n^+} \right\}. \quad (3)$$

Theorem 1 (Conditional composite exterior route). *If every input in (3) has the stated one-sided validity for the same K_n , then*

$$B_n \leq q_4(K_n).$$

Proof. The direct term is valid by assumption. The spectral term follows from (1), the trace term from (2) and (1), and the directional term from the relative Rayleigh comparison followed by (1). The maximum of valid lower bounds remains a lower bound. $\square$

Corollary 2 (Liminf closure). *For a fixed threshold $\tau > 0$, if*

$$\liminf_{n \rightarrow \infty} B_n > \tau,$$

then $q_4(K_n) \geq \tau$ for every sufficiently large n .

Proof. A strict liminf gap makes $B_n > \tau$ eventually. Apply Theorem 1 level by level. $\square$

No single route has to win eventually. Direct, trace, and directional terms may alternate indefinitely; only their admitted maximum needs a strict liminf gap. This is the alternating-route closure permitted by the composite theorem.

RH-116 supplies an exact algorithmic interpretation for the direct term. Along nested geometric memory windows, the Weyl lower is monotone in depth; the first passing depth is cost-minimal, and full-history search is complete for that certificate family. A uniform depth bound would improve cost, but is not logically required by Corollary 2.

3 Outward robustness

Different numerical assemblies cannot be fused solely because they agree to displayed digits. Suppose candidate $C_{i,n}$ has an outward transport loss $r_{i,n} \geq 0$. Its admitted value is

$$\tilde{C}_{i,n} = \max\{0, C_{i,n} - r_{i,n}\}.$$

Proposition 3 (Strict-margin outward robustness). *If the transported candidates are valid for one enclosed K_n and*

$$\liminf_{n \rightarrow \infty} \max_i \tilde{C}_{i,n} > \tau,$$

then eventual support follows. In particular, any uniform strict gate gap larger than the worst transport loss survives outward admission.

Proof. Downward transport preserves the lower-bound direction. Apply Corollary 2 to the transported maximum. $\square$

This proposition formalizes the RH-115 admission filter. RH-116 avoids the specific weak-mode interface by using one internally consistent Gram path; its numerical lower values are therefore reported as a separate self-contained route rather than silently merged with RH-115 values.

4 Minimal physical condition packets

Within the present ledger, all-level closure can arrive through any one of three compound inequalities.

“Minimal” here is ledger-relative: each row is one scalar closing condition after all algebraic factors have been eliminated. It is not a claim that no future theorem could discover a different primitive route. The trivial universal capacity bound $\Lambda_{23} \leq 1$ is available, but a sharper physical capacity estimate can greatly reduce the required volume lower.

RH-117 proves that five anchors, even together with the range $0 < \Lambda_{23} < 1$, cannot establish any of these liminf conditions by extrapolation. A new inequality linking levels is necessary.

packet	compound liminf condition	primitive physical inputs	status
direct	$D_n^- > \tau$	recent fourth-mode margin vs. tail	open
trace	$\sqrt{\Theta_n^-/\kappa_n^+}/L_n^+ > \tau$	trace lower; concentration and capacity uppers	open
directional	$(1 - \gamma_n^+)_+^4 a_n^-/L_n^+ > \tau$	frame-volume lower; gamma and capacity uppers	open

Table 1: Minimal condition packets relative to the current theorem ledger.

5 Complete finite-gate audit

The audit aligns RH-115’s admitted candidates and RH-116’s adaptive direct certificates on five scales, two sides, three thresholds, and 360 labelled records. Reconstructing the RH-115 maximum produces zero discrepancies. The two assembly paths agree on all support labels and have no false positives.

threshold	actual	direct	trace	directional	composite	adaptive
10^{-8}	115	113	113	114	114	115
10^{-6}	109	109	105	109	109	109
10^{-4}	98	98	95	98	98	98

Table 2: Full-chain support counts out of 120 updates per threshold.

Across thresholds there are 322 genuinely supported records. The admitted RH-115 composite closes 321 and misses one weak 10^{-8} record. The self-contained adaptive route closes all 322. Every fine-chain record is closed at every threshold. Within the RH-115 maximum, direct Weyl wins 189 labelled records and directional Rayleigh wins 171; the winner genuinely alternates.

6 Consequence and boundary

The route is now logically complete in conditional form. Algebraic factorization, monotone fusion, outward degradation, and finite-depth search no longer hide implicit assumptions. What remains is physical: prove a strict all-level liminf condition for at least one packet, together with same-operator outward enclosures where independent numerical paths are used.

At present the count of proved all-level physical packets is zero out of three. Therefore this paper does not close uniform Stage A, construct a Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.

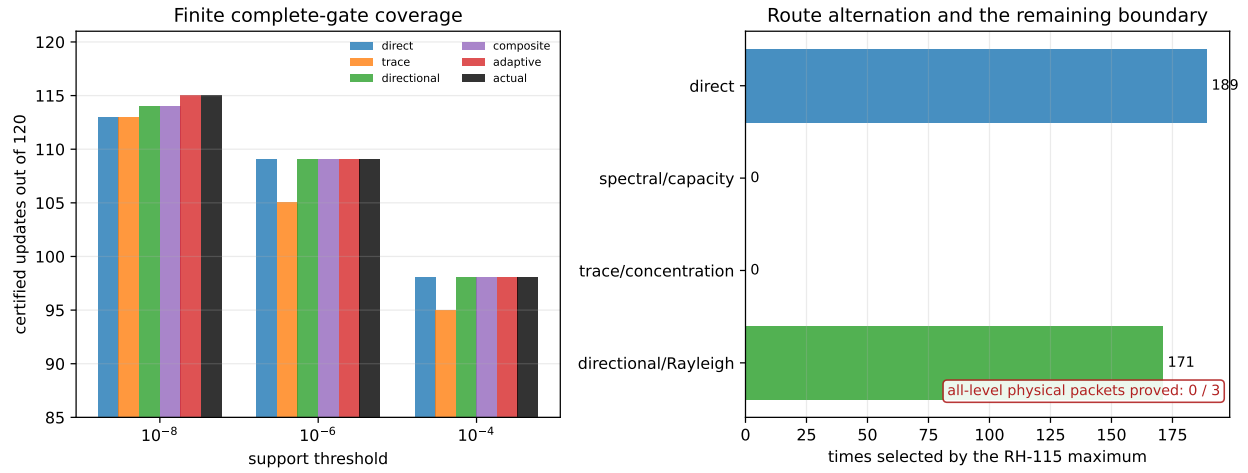


Figure 1: Left: finite support coverage. Right: route alternation and the unresolved all-level boundary.

Ramon E. Moore. *Interval Analysis*. Prentice-Hall, 1966.